

Web Music Player

Domain Model

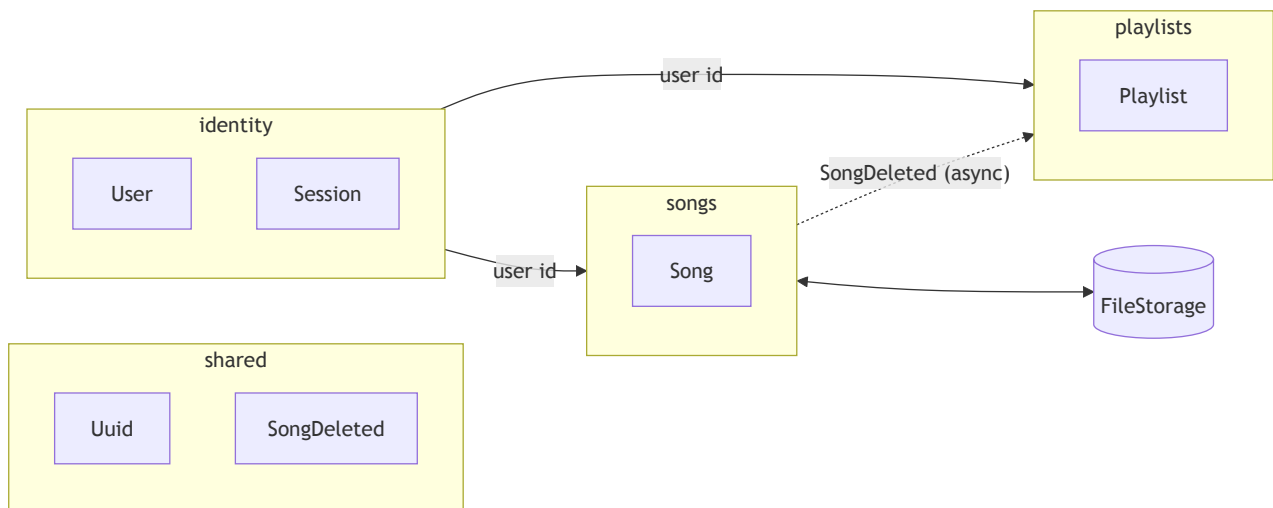
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Contents

1 Contexts	1
2 Aggregates	2
3 Rules	2
4 Ports	2
5 Domain Event	3
6 Code Layout	3

1 Contexts

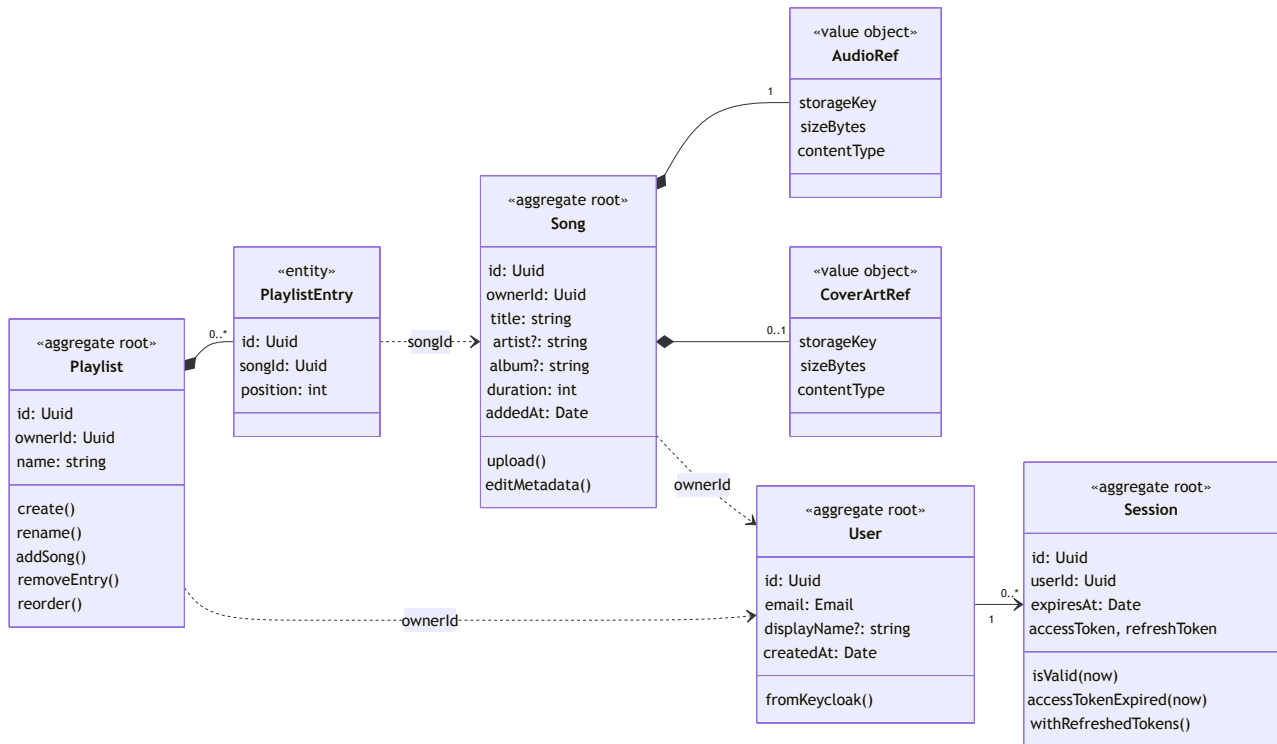
Three bounded contexts, one NestJS module each (`0002-ddd-hexagonal-backend`). Contexts reference each other by **id only**.



Context	Owns
identity	Local copy of Keycloak users, login sessions. Stories AUTH-1/2.
songs	Upload, metadata, audio + cover files, range streaming. SNG-1–3, PB-1.
playlists	Ordered song lists; a song may appear twice. PL-1–3.

Playback state (current track, position, volume) is client state in the Angular `PlayerService`, not a backend context.

2 Aggregates



Dashed arrows are id references across contexts.

3 Rules

Aggregate	Invariants
User	id is the Keycloak sub. Email is trimmed, lower-case, valid. No password stored — Keycloak owns credentials.
Session	id is the cookie value. Invalid after expiresAt. Access token refreshed 5 s before it expires. Tokens never leave the server.
Song	title not empty (falls back to the filename). duration, audio, coverArt never change. Upload must be audio/mpeg, .mp3, ≤ 20 MB, with a readable duration.
Playlist	name not empty (duplicates allowed). position is always 0..n-1. reorder must be a permutation of the current entries. The same songId may repeat — each occurrence has its own entry id.

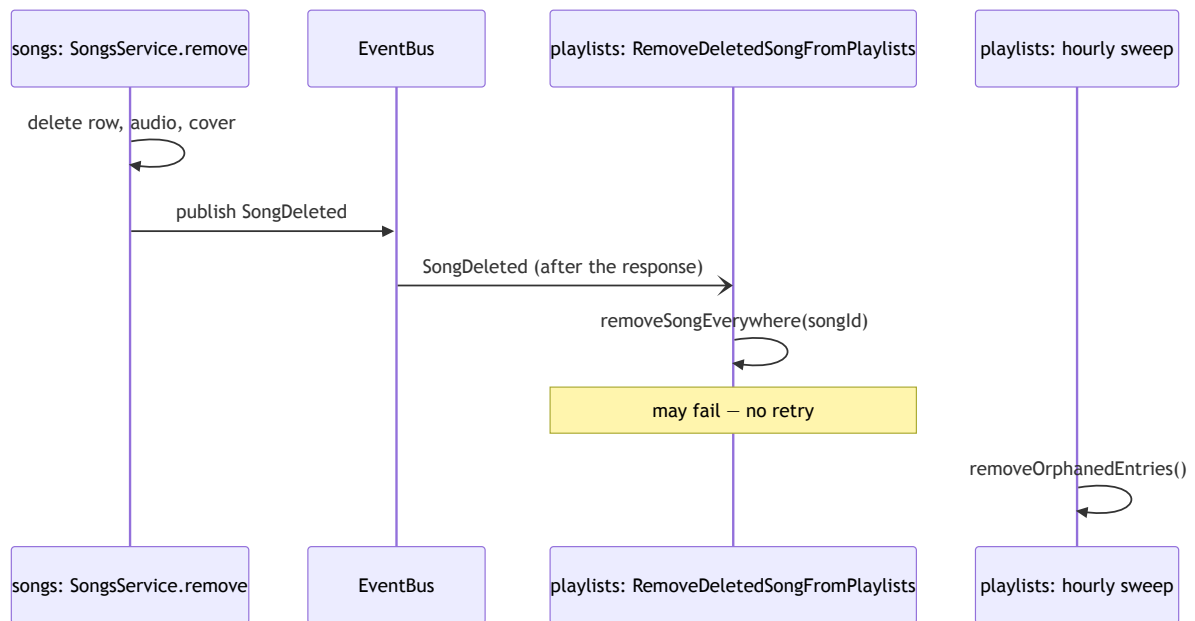
4 Ports

Port	Methods
UserRepository	save, byId
SessionRepository	save, byId, deleteById, deleteExpired
TokenVerifier	verify(accessToken) — JWKS, iss, aud, expiry
OIDCClient	authorizeUrl, exchangeCode, refresh, endSession
SongRepository	save, byId(id, ownerId), listByOwner(ownerId, sort), remove
FileStorage	put, getRange, delete, stat (0004-metadata-postgres-blob-storage-port)
Id3Reader	read(bytes)
PlaylistRepository	save, byId(id, ownerId), listByOwner, remove, removeSongEverywhere, removeOrphanedEntries

Clock, IdGenerator	shared
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5 Domain Event

`SongDeleted(songId, ownerId)` is the only event between contexts.



6 Code Layout

`backend/src/<context>/` with `domain/`, `application/`, `infrastructure/`, `http/` and `<context>.module.ts`. An ESLint rule blocks imports between contexts; shared types live in `backend/src/shared/`.